

Uniformly distributed Steiner directions need not converge

Literature-answer note for arXiv:0902.0462

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Status. `literature_already_answered` (two deterministic questions).

Source questions

Volcic [1] asks on PDF pages 12–13 whether there exist a uniformly distributed sequence of directions (u_m) and a bounded Caccioppoli, convex, compact, summable, or finite-measure set F such that its iterated Steiner symmetrals fail to converge to the centered equimeasurable ball in the appropriate metric. He expects a negative answer and makes the stronger conjecture that density alone of (u_m) suffices for convergence.

Later counterexample

Bianchi, Burchard, Gronchi, and Volcic [2] give exactly the requested example. In their Example 2.1 (PDF pages 3–4), choose $(\alpha_m) \subset (0, \pi/2)$ with

$$\sum_m \alpha_m = \infty, \quad \sum_m \alpha_m^2 < \infty, \quad \beta_m = \sum_{k=1}^m \alpha_k, \quad u_m = (\cos \beta_m, \sin \beta_m).$$

Writing $\gamma = \prod_m \cos \alpha_m > 0$, they take a planar convex body K of area smaller than a disk of diameter γ which contains a vertical unit segment. Successive symmetrizations preserve a rotating segment whose length tends to γ . If the symmetrals converged in Hausdorff distance, their limit would contain a disk of diameter γ , contradicting preservation of the smaller area of K . Hence the symmetrals do not converge.

Section 5 of the same paper (PDF page 9) makes this example uniformly distributed. Take, for instance,

$$\alpha_m = \vartheta m^{-\sigma}, \quad \frac{1}{2} < \sigma < 1, \quad 0 < \vartheta < \frac{\pi}{2}.$$

Then $\alpha_m \downarrow 0$, $m\alpha_m \rightarrow \infty$, and $\sum_m \alpha_m^2 < \infty$. The cited uniform-distribution criterion gives that (u_m) is uniformly distributed on S^1 , yet the symmetrals of the compact convex K from Example 2.1 do not converge.

Resolution and scope

Therefore the source existence question has answer **yes**, already for a bounded compact convex planar body in the Hausdorff metric. Since every uniformly distributed sequence is dense, the same example disproves the stronger density-only convergence conjecture. This is an explicit later-literature resolution, not a new result. It does not settle the source paper's separate questions about unbounded sets, nonuniform symmetric random laws, or weakened independence.

References

- [1] A. Volcic, *Random symmetrizations of measurable sets*, Calc. Var. Partial Differential Equations **40** (2011), 79–90; [arXiv:0902.0462](#).
- [2] G. Bianchi, A. Burchard, P. Gronchi, and A. Volcic, *Convergence in shape of Steiner symmetrizations*, Indiana Univ. Math. J. **61** (2012), 1695–1710; [arXiv:1206.2041](#).